

Lab Assignment 04, Tactical Design & Event Storming

Application Scenario 4: Instant Mobility Platform

Team: The Winx | Summer Semester 2026

Task 1: Tactical Design

Each bounded context from Assignment 03 has been assigned to one team member. That member is responsible for the tactical DDD design and the UML class diagram of their context, and will later own its implementation. The remaining contexts will be mocked.

1.1 Team Assignment

BC	Bounded Context	Responsible Member	Implementation
BC-01	Identity & Access	Sama Al Zou'bi	Full implementation
BC-02	Fleet Management	Priyanka Gupta	Full implementation
BC-03	Booking	Rowena Pagayanan	Full implementation
BC-04	Payment	Marianne Nosseir	Full implementation
BC-05	Rating	Maede Eskandari	Full implementation

1.2 DDD Building Block Reference

The following UML stereotypes are used in all class diagrams below, following the notation from Paper 01 (Table 1) and Paper 02 (Table 3) provided in the lab materials.

Stereotype	Fill	Border	Role in design
<<AggregateRoot>>	Amber	Thick solid	Entry point to an aggregate. All access to internal objects goes through it. Enforces aggregate invariants.
<<Entity>>	Light yellow	Solid	Has a unique identity. State changes over time. Tracked by ID rather than attribute values.

<<ValueObject>>	Green	Dashed	No identity. Defined entirely by its attributes. Immutable, replaced, never mutated.
<<Service>>	Blue	Solid	Stateless. Contains domain logic that does not naturally belong to an entity or value object.
<<Repository>>	Purple	Solid	Provides persistence operations for aggregate roots. Hides storage implementation details.
<<DomainEvent>>	Orange	Solid	Records that something significant happened in the domain. Immutable. Published after state change.

1.3 Tactical Design per Bounded Context

BC-01 Identity & Access, Sama Al Zou'bi

This context owns all authentication and account management logic. The two aggregate roots (UserAccount and ProviderAccount) encapsulate credential validation and registration. Value Objects enforce immutability of credentials and contact data. No booking or fleet logic exists here.

Aggregates

Aggregate Root	Members	Key Invariants
UserAccount	PersonallInfo (Embedded), email: String, passwordHash: String	Email and Password are not modelled as custom classes in Java; they are represented as primitive String columns (email and passwordHash)
ProviderAccount	CompanyInfo (Embedded), email: String, passwordHash: String, phoneNumber: String.	In ProviderAccount, phoneNumber is defined as a field on the entity itself, not within CompanyInfo.

Entities

Entity	Identity Attribute	Reason, why not a Value Object?
UserAccount	Id : Long	Has a persistent identity (tracked by id). Its state changes over its lifetime (lifecycle transitions: PENDING $\rightarrow$ ACTIVE $\rightarrow$ DEACTIVATED).
ProviderAccount	Id : Long	Has a persistent identity (tracked by id). Its state changes over its lifetime (lifecycle transitions: PENDING $\rightarrow$ ACTIVE $\rightarrow$ DEACTIVATED).

Value Objects

Value Object	Attributes	Why immutable / no identity?
PersonallInfo	name, dateOfBirth, phoneNumber	Groups user personal data. Immutable snapshot, replaced, not mutated.
CompanyInfo	CompanyName : String, contactName : String	Groups provider business data. Immutable snapshot.
AuthToken	value:String, expiresAt:DateTime	Immutable session token returned on successful authentication.

Domain Services

Service	Key Operations
AuthenticationService	authenticateUser(email, rawPassword): AuthenticatedUser authenticateProvider(email, rawPassword): AuthenticatedProvider validateToken(token): Optional<PrincipalRef>
RegistrationService	registerUser(name, email, rawPassword, phoneNumber, dateOfBirth): UserAccount registerProvider(companyName, contactName, email, rawPassword, phoneNumber): ProviderAccount

Repositories

Repository	Methods
UserAccountRepository	findByEmail(email), existsByEmail(email) alongside default Spring Data CRUD operations.
ProviderAccountRepository	findByEmail(email), existsByEmail(email) alongside default Spring Data CRUD operations.

Domain Events

Domain Event	Trigger Condition
UserRegistered	RegistrationService successfully creates a UserAccount
ProviderRegistered	RegistrationService successfully creates a ProviderAccount
UserAuthenticated	AuthenticationService validates credentials and issues a token

PlantUML Source:

```

@startuml
    BC-01: Identity & Access
    skinparam classBackgroundColor<<AggregateRoot>> #FFF3CD
    skinparam classBackgroundColor<<ValueObject>> #D4EDDA
    skinparam classBackgroundColor<<Service>> #CCE5FF
    skinparam classBackgroundColor<<Repository>> #E2D9F3
    skinparam classBackgroundColor<<DomainEvent>> #FFE0B2

```

```
classBorderColor<<AggregateRoot>> #856404 skinparam
classBorderThickness<<AggregateRoot>> 2 skinparam
classBorderStyle<<ValueObject>> dashed

class UserAccount <<AggregateRoot>> {
  + userId : Long
  + registeredAt : DateTime
  + status : AccountStatus
  --
  + authenticate(pwd) : AuthToken
  + deactivate() : void
}
class ProviderAccount <<AggregateRoot>> {
  + providerId : Long
  + registeredAt : DateTime
  + status : AccountStatus
  --
  + authenticate(pwd) : AuthToken
}
class PersonalInfo <<ValueObject>> {
  + name : String +
  dateOfBirth : Date
}
class CompanyInfo <<ValueObject>> {
  + companyName : String
  + contactName : String
}
class Email <<ValueObject>> {
  + value : String
  --
  + validate() : boolean
}
class Password <<ValueObject>> {
  + hash : String
  --
  + matches(plain) : boolean
}
class PhoneNumber <<ValueObject>> { + value : String }
class AuthToken <<ValueObject>> {
  + value : String +
  expiresAt : DateTime
}
class AuthenticationService <<Service>> {
  + authenticate(email,pwd) : AuthToken +
  validateToken(token) : Principal
}
class RegistrationService <<Service>> { +
  registerUser(...) : UserAccount +
  registerProvider(...) : ProviderAccount
}
interface UserRepository <<Repository>> {
  + findById(id) : UserAccount
  + findByEmail(e) : UserAccount
  + save(u) : void
}
interface ProviderRepository <<Repository>> {
  + findById(id) : ProviderAccount
  + findByEmail(e) : ProviderAccount
  + save(p) : void
}
class UserRegistered <<DomainEvent>> { + userId:Long + email:String } class
class ProviderRegistered <<DomainEvent>> { + providerId:Long + email:String }
```

```

UserAccount *-- PersonalInfo
UserAccount *-- Email
UserAccount *-- Password
PersonalInfo *-- PhoneNumber
ProviderAccount *-- CompanyInfo
ProviderAccount *-- Email
ProviderAccount *-- Password
CompanyInfo *-- PhoneNumber
AuthenticationService ..> UserRepository
AuthenticationService ..> ProviderRepository
RegistrationService ..> UserRepository
RegistrationService ..> ProviderRepository
RegistrationService ..> UserRegistered : publishes
RegistrationService ..> ProviderRegistered : publishes
@enduml

```

BC-02 Fleet Management , Priyanka Gupta

This context is the single source of truth for vehicles. The Vehicle aggregate encapsulates location, pricing, and restriction data as embedded value objects so that vehicle integrity can be enforced as a unit. Provider identity comes from BC-01 via ACL, only the providerId reference is stored here.

Aggregates

Aggregate Root	Members	Key Invariants
Vehicle	Vehicle Location, PricingPolicy, UsageRestrictions	pricePerUnit must be > 0. A Vehicle cannot be deleted while its status is BOOKED. Restrictions are optional but validated as non-negative when present.

Entities

Entity	Identity Attribute	Reason, why not a Value Object?
Vehicle	id : Long	Has persistent identity (vehicleId). Status, location, and pricing change over its lifetime.

Value Objects

Value Object	Attributes	Why immutable / no identity?
VehicleLocation	latitude:Double, longitude:Double	Immutable GPS snapshot. Replaced entirely on each location update.
PricingPolicy	pricePerUnit:BigDecimal, billingModel:String	The codebase stores billingModel as a standard string in the database column.
UsageRestrictions	maxDurationMins, maxKm, minAge, maxPersons (all nullable Integer)	Immutable. Groups all optional restriction rules. Null means no restriction.

VehicleType	E_SCOOTER BICYCLE E_BIKE E_CAR	Enum-backed value object. Fixed vocabulary, cannot be extended at runtime.
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Domain Services

Service	Key Operations
VehicleRegistrationService	createVehicle(providerId, type, desc, pricing, restrictions):Vehicle
FleetStatusService	getFleetOverview(providerId): List<Vehicle> updateLocation(vehicleId, latitude, longitude): Vehicle
VehicleAvailabilityService	findAvailableNear(lat, lon, radiusKm):List<Vehicle>, findById(id):Vehicle

Repositories

Repository	Methods
VehicleRepository	findById(id), findByProviderId(pid), findAvailableNear(lat,lon,radius), save(v), delete(id)

Domain Events

Domain Event	Trigger Condition
VehicleCreated	VehicleRegistrationService successfully registers a new vehicle
VehicleStatusUpdated	Booking context sends status-change command (AVAILABLE ↔ BOOKED)
VehicleLocationUpdated	FleetStatusService records new GPS coordinates
VehicleDeleted	Provider deletes a vehicle that is currently AVAILABLE

PlantUML Source:

```
@startuml BC-02: Fleet Management
skinparam classBackgroundColor<<AggregateRoot>> #FFF3CD
skinparam classBackgroundColor<<ValueObject>> #D4EDDA
skinparam classBackgroundColor<<Service>> #CCE5FF skinparam
classBackgroundColor<<Repository>> #E2D9F3 skinparam
classBackgroundColor<<DomainEvent>> #FFE0B2 skinparam
classBorderThickness<<AggregateRoot>> 2 skinparam
classBorderStyle<<ValueObject>> dashed

class Vehicle <<AggregateRoot>> {
    + vehicleId : Long
    + providerId : Long
    + description : String
    + status : VehicleStatus
    --
    + markBooked() : void
}
```

```
+ markAvailable() : void +
updateLocation(loc) : void
}
class VehicleLocation <<ValueObject>> {
+ latitude : Double
+ longitude : Double
}
class PricingPolicy <<ValueObject>> {
+ pricePerUnit : BigDecimal +
billingModel : BillingModel
}
class UsageRestrictions <<ValueObject>> {
+ maxDurationMins : Integer
+ maxKilometers : Integer
+ minAge : Integer +
maxPersons : Integer
}
enum VehicleType { E_SCOOTER / BICYCLE / E_BIKE / E_CAR }
enum VehicleStatus { AVAILABLE / BOOKED } enum BillingModel
{ PER_HOUR / PER_KILOMETER } class
VehicleRegistrationService <<Service>> {
+ createVehicle(...) : Vehicle
}
class FleetStatusService <<Service>> {
+ getFleetOverview(pid) : List +
updateLocation(vid, loc) : void
}
interface VehicleRepository <<Repository>> {
+ findById(id) : Vehicle
+ findByProviderId(pid) : List
+ findAvailableNear(lat,lon,r) : List
+ save(v) : void
+ delete(id) : void
}
class VehicleCreated <<DomainEvent>> { + vehicleId:Long + providerId:Long }
class VehicleStatusUpdated <<DomainEvent>> { + vehicleId:Long + newStatus:VehicleStatus }

Vehicle *-- VehicleLocation
Vehicle *-- PricingPolicy
Vehicle *-- UsageRestrictions
Vehicle --> VehicleType
Vehicle --> VehicleStatus
PricingPolicy --> BillingModel
VehicleRepository ..> Vehicle
VehicleRegistrationService ..> VehicleRepository
FleetStatusService ..> VehicleRepository
VehicleRegistrationService ..> VehicleCreated : publishes
FleetStatusService ..> VehicleStatusUpdated : publishes
@enduml
```

BC-03 Booking , Rowena Pagayanan

The Booking context is the operational heart of the platform. The Booking aggregate root enforces all ride lifecycle invariants. A VehicleSnapshot value object captures pricing at booking-time to prevent retroactive pricing changes from affecting in-progress rides. Cost computation and payment triggering are handled by domain services.

Aggregates

Aggregate Root	Members	Key Invariants
Booking	RideLocation (start/end), TimeInterval, VehicleSnapshot, RideSummary	endTime must be after startTime. A COMPLETED booking cannot transition to CANCELLED. totalCost must be computed before status reaches COMPLETED. Only one active booking per User at a time.

Entities

Entity	Identity Attribute	Reason, why not a Value Object?
Booking	Id : Long	Has persistent identity (bookingId). Status transitions define its entire lifecycle: ACTIVE → COMPLETED or CANCELLED.

Value Objects

Value Object	Attributes	Why immutable / no identity?
RideLocation	latitude:Double, longitude:Double	Immutable GPS snapshot captured at the start and end of the ride. Modeled as a Java record, so it has no identity and is structurally immutable.
TimeInterval	start:LocalDateTime, end:LocalDateTime	Immutable range representing booking times. Modeled as a Java record (structurally immutable). The end time is null during active rides, and replaced when the ride is completed.
VehicleSnapshot	vehicleId, type, pricePerUnit, billingModel	Immutable snapshot of pricing at booking time. Decouples Booking from future vehicle price changes.
RideSummary	distanceKm:Double, totalCost:BigDecimal	Immutable result of cost computation. Null until ride is COMPLETED.

Domain Services

Service	Key Operations
VehicleSearchService	findAvailableNear(userId, lat, lon, filters):List<VehicleSnapshot>
BookingService	endBooking(bookingId, endLatitude, endLongitude, paymentMethod): BookingEndResult
CostCalculationService	computeCost(snapshot: VehicleSnapshot, start: LocalDateTime, end: LocalDateTime, distanceKm: double): BigDecimal
RestrictionValidator	validate(userId:UserRef, vehicle:VehicleSnapshot):void, throws if age/time/km restrictions violated

Repositories

Repository	Methods
BookingRepository	findById(id), findByUserId(uid), findActiveByUserId(uid), findByVehicleId(vid), save(b)

Domain Events

Domain Event	Trigger Condition
BookingCreated	BookingService creates a new ACTIVE booking; vehicle status updated to BOOKED
BookingCompleted	BookingService ends ride, cost computed, payment trigger sent
BookingCancelled	BookingService cancels an ACTIVE booking; vehicle released
PaymentTriggered	Published after BookingCompleted; consumed by Payment context

PlantUML Source:

```
@startuml BC-03: Booking
skinparam classBackgroundColor<<AggregateRoot>> #FFF3CD
skinparam classBackgroundColor<<ValueObject>> #D4EDDA
skinparam classBackgroundColor<<Service>> #CCE5FF skinparam
classBackgroundColor<<Repository>> #E2D9F3 skinparam
classBackgroundColor<<DomainEvent>> #FFE0B2 skinparam
classBorderThickness<<AggregateRoot>> 2 skinparam
classBorderStyle<<ValueObject>> dashed
```

```
class Booking <<AggregateRoot>> {
    + bookingId : Long
    + userId : Long
    + status : BookingStatus
    --
    + complete(endLoc, distKm, cost) : void
    + cancel() : void
}
class RideLocation <<ValueObject>> {
    + latitude : Double
    + longitude : Double
}
class TimeInterval <<ValueObject>> {
    + startTime : DateTime
    + endTime : DateTime
}
class VehicleSnapshot <<ValueObject>> {
    + vehicleId : Long
    + type : VehicleType
    + pricePerUnit : BigDecimal
    + billingModel : BillingModel
}
class RideSummary <<ValueObject>> {
    + distanceKm : Double
    + totalCost : BigDecimal
}
```

```
enum BookingStatus { ACTIVE / COMPLETED / CANCELLED }
class BookingService <<Service>> { +
createBooking(uid, vid) : Booking
+ endBooking(bid, loc) : Booking
+ cancelBooking(bid) : Booking
}
class CostCalculationService <<Service>> { +
computeCost(snap, interval, km) : BigDecimal
}
class VehicleSearchService <<Service>> { +
findAvailableNear(lat,lon,filters) : List
}
class RestrictionValidator <<Service>> {
+ validate(userId, snapshot) : void
}
interface BookingRepository <<Repository>> {
+ findById(id) : Booking
+ findByUserId(uid) : List
+ findActiveByUserId(uid) : Booking
+ save(b) : void
}
class BookingCreated <<DomainEvent>> { + bookingId:Long + userId:Long + vehicleId:Long }
class BookingCompleted <<DomainEvent>> { + bookingId:Long + totalCost:BigDecimal } class
BookingCancelled <<DomainEvent>> { + bookingId:Long + vehicleId:Long }
```

```

class PaymentTriggered <<DomainEvent>> { + bookingId:Long + amount:BigDecimal + userId:Long
}

Booking *-- RideLocation
Booking *-- TimeInterval
Booking *-- VehicleSnapshot
Booking *-- RideSummary
Booking --> BookingStatus
BookingService ..> BookingRepository
BookingService ..> CostCalculationService
BookingService ..> RestrictionValidator
BookingService ..> BookingCreated : publishes
BookingService ..> BookingCompleted : publishes
BookingService ..> BookingCancelled : publishes
BookingService ..> PaymentTriggered : publishes @enduml

```

BC-04 Payment , Marianne Nosseir

The Payment context is triggered by the PaymentTriggered domain event from the Booking context. It is a leaf context with no outward dependencies. The Payment aggregate enforces financial consistency: amount must match the booking total, and status transitions are strictly controlled (PENDING → PAID or FAILED only).

Aggregates

Aggregate Root	Members	Key Invariants
Payment	Money, PaymentMethod, PaymentResult	Amount must equal the bookingId's totalCost received from the event. Status can only transition PENDING → PAID or PENDING → FAILED, never backwards. A PAID payment cannot be modified.

Entities

Entity	Identity Attribute	Reason, why not a Value Object?
Payment	Id : Long	Has persistent identity (paymentId). Status transitions define its lifecycle.

Value Objects

Value Object	Attributes	Why immutable / no identity?
Money	amount:BigDecimal, currency:String	Immutable monetary amount. Enforces non-negative values. Currency code validated (ISO 4217).
PaymentMethod	type:String, maskedReference:String	Immutable reference to the payment instrument used. Stored masked for PCI compliance.
PaymentResult	status:PaymentStatus, paidAt:DateTime, failureReason:String	Immutable outcome of payment processing. Null paidAt when FAILED or PENDING.

Domain Services

Service	Key Operations
PaymentProcessingService	<code>initiatePayment(bookingId, amount, currency, userId, paymentMethodType, paymentMethodMaskedRef): Payment</code>
PaymentGatewayAdapter	<code>charge(money, method):PaymentResult , infrastructure service;</code> adapts to external payment provider API

Repositories

Repository	Methods
PaymentRepository	<code>findById(id), findByIdByBookingId(bid), save(p)</code>

Domain Events

Domain Event	Trigger Condition
PaymentInitiated	PaymentProcessingService creates a new PENDING Payment in response to PaymentTriggered event
PaymentSucceeded	PaymentGatewayAdapter returns successful charge; status set to PAID
PaymentFailed	PaymentGatewayAdapter returns failure; status set to FAILED with reason

PlantUML Source: @startuml

```

BC-04: Payment
skinparam classBackgroundColor<<AggregateRoot>> #FFF3CD
skinparam classBackgroundColor<<ValueObject>> #D4EDDA
skinparam classBackgroundColor<<Service>> #CCE5FF skinparam
classBackgroundColor<<Repository>> #E2D9F3 skinparam
classBackgroundColor<<DomainEvent>> #FFE0B2 skinparam
classBorderThickness<<AggregateRoot>> 2 skinparam
classBorderStyle<<ValueObject>> dashed

```

```

class Payment <<AggregateRoot>> {
    + paymentId : Long
    + bookingId : Long
    + userId : Long
    --
    + markPaid(paidAt) : void
    + markFailed(reason) : void
}
class Money <<ValueObject>> {
    + amount : BigDecimal
    + currency : String
}
class PaymentMethod <<ValueObject>> {
    + type : String +
    maskedReference : String
}
class PaymentResult <<ValueObject>> {
    + status : PaymentStatus
}

```

```

+ paidAt : DateTime
+ failureReason : String
}
enum PaymentStatus { PENDING / PAID / FAILED }
class PaymentProcessingService <<Service>> {
+ initiatePayment(bookingId,amount,uid) : Payment
+ retryPayment(paymentId) : Payment
}
class PaymentGatewayAdapter <<Service>> {
+ charge(money, method) : PaymentResult
}
interface PaymentRepository <<Repository>> {
+ findById(id) : Payment
+ findByBookingId(bid) : Payment
+ save(p) : void
}
class PaymentInitiated <<DomainEvent>> { + paymentId:Long + bookingId:Long + amount:BigDecimal
}
class PaymentSucceeded <<DomainEvent>> { + paymentId:Long + paidAt:DateTime } class
PaymentFailed <<DomainEvent>> { + paymentId:Long + reason:String }

Payment *-- Money
Payment *-- PaymentMethod
Payment *-- PaymentResult
PaymentResult --> PaymentStatus
PaymentProcessingService ..> PaymentRepository
PaymentProcessingService ..> PaymentGatewayAdapter
PaymentProcessingService ..> PaymentInitiated : publishes
PaymentProcessingService ..> PaymentSucceeded : publishes
PaymentProcessingService ..> PaymentFailed : publishes @enduml

```

BC-05 Rating , Maede Eskandari

The Rating context allows users to submit feedback after completing a ride. The aggregate enforces the one-rating-per-booking invariant and validates score ranges. Vehicle and Provider data arrive as lightweight snapshots from Fleet Management via ACL. No full fleet entities exist in this context.

Aggregates

Aggregate Root	Members	Key Invariants
Rating	Review, RatingTarget	vehicleScore and providerScore must be 1–5. Exactly one Rating may exist per bookingId. Rating can only be submitted if the referenced booking is COMPLETED. Once submitted, a Rating is immutable.

Entities

Entity	Identity Attribute	Reason, why not a Value Object?
Rating	id : Long	Has persistent identity (ratingId). Represents a unique user feedback record tied to one booking.

Value Objects

Value Object	Attributes	Why immutable / no identity?
Review	vehicleScore:Score, providerScore:Score, comment:String	vehicleScore and providerScore are standard Integers “Constraints (1–5 score validation) are enforced in the domain service layer.”
RatingTarget	vehicleId:Long, providerId:Long, bookingId:Long	Immutable reference triplet. Identifies what and why is being rated.

Domain Services

Service	Key Operations
RatingSubmissionService	submitRating(bookingId, userId, vehicleScore, providerScore, comment):Rating, validates booking completion and duplicate check
RatingQueryService	getVehicleRatings(vehicleId):List<Rating>, getProviderRatings(providerId):List<Rating>, getAverageScore(vehicleId):Double

Repositories

Repository	Methods
RatingRepository	findById(id), findByBookingId(bid), findByVehicleId(vid), findByProviderId(pid), existsByBookingId(bid):boolean, save(r)

Domain Events

Domain Event	Trigger Condition
RatingSubmitted	RatingSubmissionService successfully creates and persists a new Rating

PlantUML Source: @startuml

```

BC-05: Rating
skinparam classBackgroundColor<<AggregateRoot>> #FFF3CD
skinparam classBackgroundColor<<ValueObject>> #D4EDDA
skinparam classBackgroundColor<<Service>> #CCE5FF skinparam
classBackgroundColor<<Repository>> #E2D9F3 skinparam
classBackgroundColor<<DomainEvent>> #FFE0B2 skinparam
classBorderThickness<<AggregateRoot>> 2 skinparam
classBorderStyle<<ValueObject>> dashed

class Rating <<AggregateRoot>> {
    + ratingId : Long
    + userId : Long
    + createdAt : DateTime
}
class Review <<ValueObject>> {
    + vehicleScore : Score
    + providerScore : Score

```

```
+ comment : String
}
class Score <<ValueObject>> {
  + value : Integer
  --
  + {static} of(v) : Score
}
class RatingTarget <<ValueObject>> {
  + vehicleId : Long
  + providerId : Long
  + bookingId : Long
}
class RatingSubmissionService <<Service>> {
  + submitRating(bid,uid,vs,ps,c) : Rating
}
class RatingQueryService <<Service>> {
  + getVehicleRatings(vid) : List
  + getAverageScore(vid) : Double
}
interface RatingRepository <<Repository>> {
  + findByBookingId(bid) : Rating
  + findByVehicleId(vid) : List
  + existsByBookingId(bid) : boolean
  + save(r) : void
}
class RatingSubmitted <<DomainEvent>> {
  + ratingId : Long
  + bookingId : Long
  + vehicleScore : Integer
  + providerScore : Integer
}

Rating *-- Review
Rating *-- RatingTarget
Review *-- Score
RatingSubmissionService ..> RatingRepository
RatingQueryService ..> RatingRepository
RatingSubmissionService ..> RatingSubmitted : publishes @enduml
```

Task 2: Event Storming

The team chose Event Storming to model the domain of the Instant Mobility Platform. Event Storming maps the system as a timeline of Domain Events (things that happened), with the Commands that trigger them, the Aggregates that process them, and the Policies that react to them.

Colour convention used in the Event Storming board:

- Orange , Domain Event (something significant happened, named in past tense)
- Blue , Command (an intention to change state, e.g. from a user or policy)
- Yellow , Aggregate (the DDD aggregate root that processes the command)
- Purple , Policy / Reaction (a rule: 'when event X happens, execute command Y')
- Green , Read Model (a view or query result used to support a decision)
- Tan , Actor (the person or system that issues a command)

2.1 Event Flow Summary

Phase	Actor	Command	Aggregate	Domain Event
Registration				
Registration	User	Register User	UserAccount	UserRegistered
Registration	Provider	Register Provider	ProviderAccount	ProviderRegistered
Registration	User	Login	UserAccount	UserAuthenticated
Vehicle Management				
Vehicle Management	Provider	Create Vehicle	Vehicle	VehicleCreated
Vehicle Management	Provider	Update Vehicle	Vehicle	VehicleUpdated
Vehicle Management	Provider	Delete Vehicle	Vehicle	VehicleDeleted
Search & Booking				
Search & Booking	User	Search Vehicles	[Read Model]	VehiclesFound
Search & Booking	User	Book Vehicle	Booking	BookingCreated
		Policy reaction:	Mark Vehicle BOOKED	
Search & Booking	—	Update Status	Vehicle	StatusUpdated
Ride & Completion				
Ride & Completion	User	End Booking	Booking	BookingCompleted
		Policy reaction:	Compute Cost	
Ride & Completion	—	Compute Cost	Booking	CostComputed

		Policy reaction:	Trigger Payment	
Ride & Completion	—	Process Payment	Payment	PaymentSucceeded
		Policy reaction:	Release Vehicle	
Ride & Completion	User	Cancel Booking	Booking	BookingCancelled
		Policy reaction:	Release Vehicle	
Rating				
Rating	—	[Allow Rating]	[Policy]	RatingAllowed
		Policy reaction:	After BookingCompleted+PaymentSucceeded	
Rating	User	Submit Rating	Rating	RatingSubmitted

2.2 Event Storming Narrative

Phase 1, Registration: The process begins when a User or Provider registers an account (UserRegistered / ProviderRegistered). Login produces a UserAuthenticated event that provides the session token used in all subsequent operations.

Phase 2, Vehicle Management: Providers create, update, and delete vehicles in their fleet. VehicleCreated is the pivotal event, without it, no bookings are possible. VehicleStatusUpdated is raised both from this context (manual updates) and reactively from the Booking context.

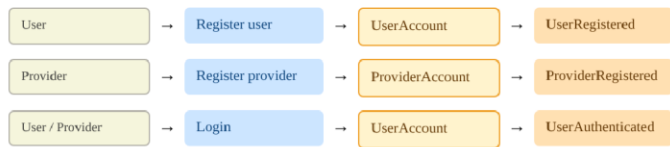
Phase 3, Search & Booking: A User searches for available vehicles (producing a VehiclesFound read model). Booking a vehicle produces BookingCreated and triggers the policy 'Mark Vehicle BOOKED' which sends a status-update command back to Fleet Management.

Phase 4, Ride & Completion: Ending a booking triggers cost computation (CostComputed), which in turn triggers the PaymentTriggered event. The Payment context processes this asynchronously, PaymentSucceeded releases the vehicle (AVAILABLE again) via another policy. If the user cancels, BookingCancelled immediately releases the vehicle.

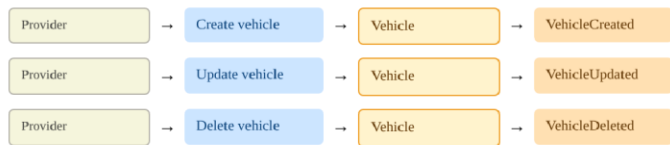
Phase 5, Rating: Only after both BookingCompleted and PaymentSucceeded have fired can the Rating context accept a rating submission. The policy guards this gate. RatingSubmitted is the final event in the happy path.

Command Aggregate Domain event Policy / reaction Read model Actor

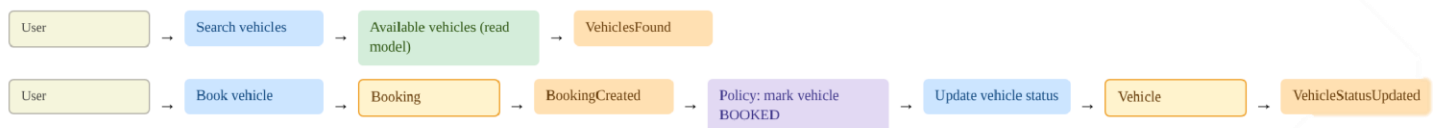
PHASE 1 — REGISTRATION (BC-01: IDENTITY & ACCESS)



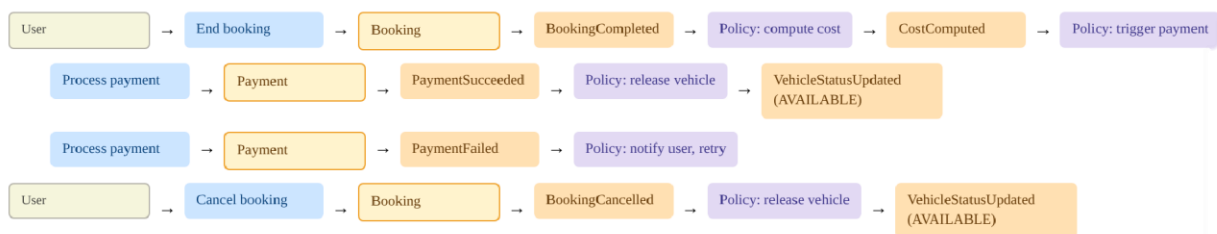
PHASE 2 — VEHICLE MANAGEMENT (BC-02: FLEET MANAGEMENT)



PHASE 3 — SEARCH & BOOKING (BC-03: BOOKING)



PHASE 4 — RIDE & COMPLETION (BC-03 + BC-04: BOOKING + PAYMENT)



PHASE 5 — RATING (BC-05: RATING)

